



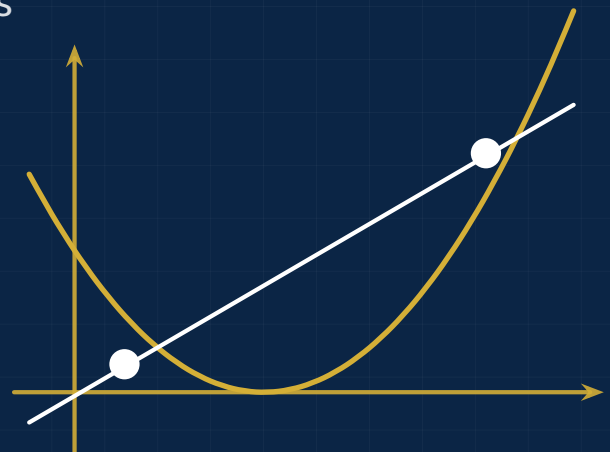
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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 4

Linear Graphs

$$y = mx + c$$

m is the steepness, c is where it crosses the y -axis. Read them, draw them, find them — and flip m for a perpendicular.



5

VARIANTS

7

PAST-PAPER PARTS

58

PRACTICE QUESTIONS

YOU WILL MASTER

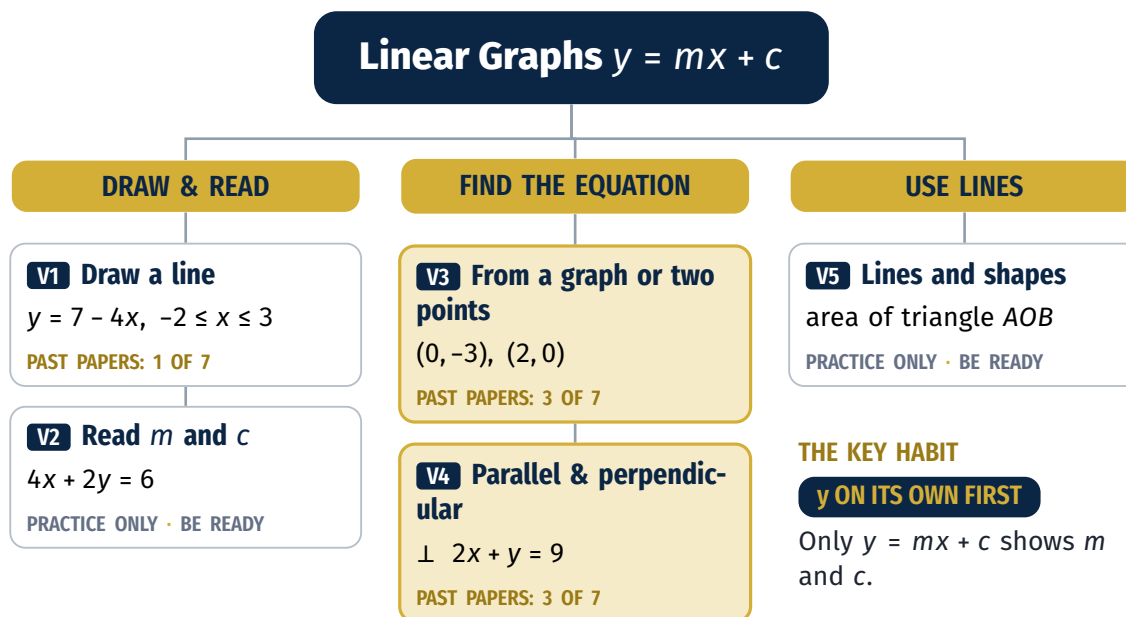
V1 Draw a line V2 Read m and c V3 Equation from a graph

V4 Parallel & perpendicular V5 Lines and shapes

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Linear graphs at a glance



Your notes


★ How often does each variant come up?

7 past-paper questions (Oct 2020 → Nov 2024); 58 practice parts — grouped by type. Harder line work is in Topic 2 (Coordinate Geometry).

	PAST PAPERS (OUT OF 7)		PRACTICE PARTS	
V1 Draw a line	● ○ ○ ○ ○ ○ ○	1	7	
V2 Read m and c	○ ○ ○ ○ ○ ○ ○	0	8	
V3 From a graph, two points	● ● ● ○ ○ ○ ○	3	11	
V4 Parallel, perpendicular	● ● ● ○ ○ ○ ○	3	18	
V5 Lines and shapes	○ ○ ○ ○ ○ ○ ○	0	14	

WHAT THIS TELLS YOU Short, early questions (Q3–Q18, 2–4 marks): draw a line, read its equation off a grid, or a parallel/perpendicular line. Quick marks once $y = mx + c$ is automatic.

BEFORE YOU START

The big idea

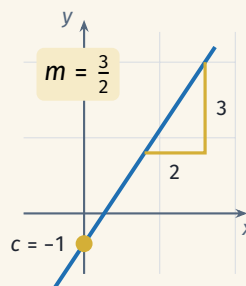


$y = mx + c$ tells you two things.

m — the gradient: how far up for 1 across (negative: downhill).

c — the y -intercept: where the line crosses the y -axis, at $(0, c)$.

Every point (x, y) on the line makes the equation true.



Four facts

Gradient

$$m = \frac{\text{rise}}{\text{run}}$$

Up is +, down is -.

 y -intercept

$$(0, c)$$

Put $x = 0$.

 x -intercept

$$y = 0$$

Solve for x .

On the line?

substitute

Both sides equal:
yes.

Related lines

Parallel

same m

$$y = 2 - 5x \text{ and } y = 6 - 5x.$$

Perpendicular

$$m \rightarrow -\frac{1}{m}$$

$$-2 \rightarrow \frac{1}{2}.$$

Rearranged

$$5y + 2x = 7$$

$$y = -\frac{2}{5}x + \frac{7}{5} \text{ first.}$$

**CALCULATOR FIRST, THEN CHECK**

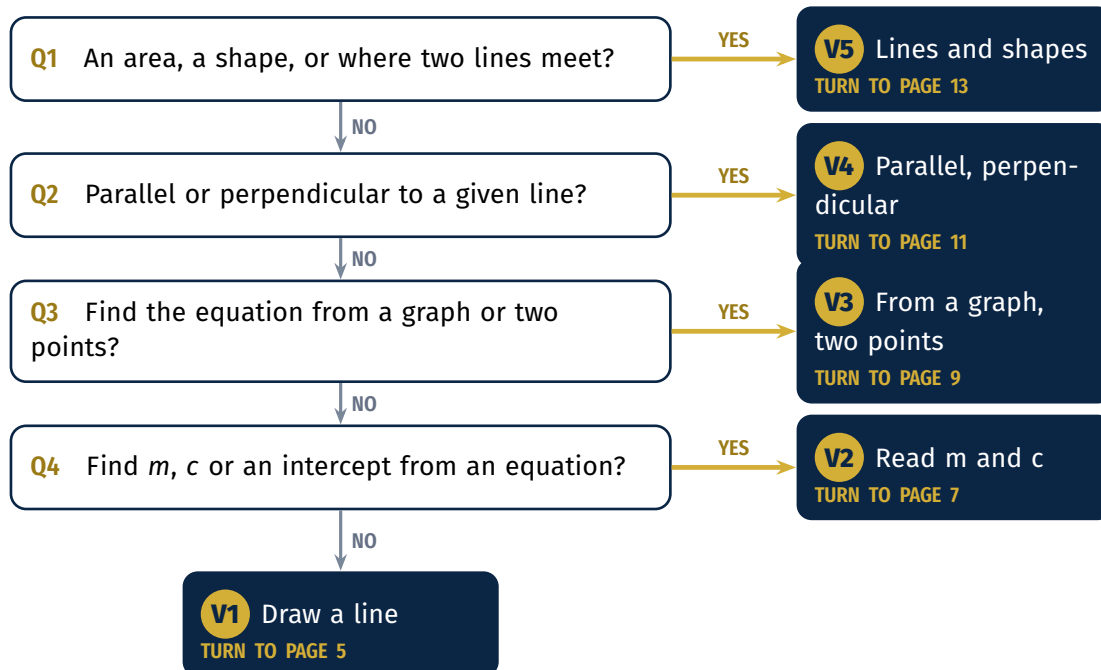
Calculator table mode ($f(x) = 7 - 4x$, start -2 , end 3 , step 1) gives the whole table in one go — then plot.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Draw a Line from its Equation

Table of values, plot every point, join with one ruled line.

SEEN IN PAST PAPERS

● ● ● ● ● ● ● 1/7

PRACTICE QUESTIONS: 7



Your notes

Came up in: Oct/Nov 2020 P1H Q3

? What is it?

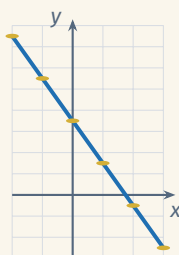
In plain words

Make a **table**: every x in the range, work out y . Plot the points — they must lie in a **straight line**.

Rule **one line** through all of them, across the whole range asked.

IT LOOKS LIKE

$$y = 7 - 4x \text{ for } -2 \leq x \leq 3$$



x	-2	-1	0	1	2	3
y	15	11	7	3	-1	-5

How do I do it?

1 Table

Every whole x in the range.

2 Work out y

Careful with negatives: $7 - 4(-2) = 15$.

3 Plot

Small crosses, exactly on the grid.

4 Join

One ruled straight line, end to end.

TABLE, PLOT, RULE A point off the line = a slip in the table.

V1 Draw a Line from its Equation

VARIANT 1 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"Draw the graph of $y = 7 - 4x$ for values of x from -2 to 3 " → **Table $x = -2, \dots, 3$; plot; rule**

"Complete the table of values" → **Substitute each x**

A point that is not in line → **Recheck that y value**

WORKED EXAMPLE **PAST PAPER**

Oct/Nov 2020 · Paper 1H · Q3 · 3 marks

On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3 .

STEP 1 $x : -2, -1, 0, 1, 2, 3$ ◀ the range

STEP 2 $y : 15, 11, 7, 3, -1, -5$ ◀ $7 - 4x$ each time

STEP 3 $(-2, 15), (-1, 11), \dots, (3, -5)$ ◀ plot

STEP 4 One ruled line from $(-2, 15)$ to $(3, -5)$ ◀ join

ANSWER straight line through $(-2, 15)$ and $(3, -5)$ ✓ $m = -4$: down 4 for each 1 across ✓

**WATCH OUT**

$$\times 7 - 4(-2) = -1$$

$$\checkmark 7 - 4(-2) = 7 + 8 = 15$$

Minus times minus is plus.

**EXAM TIP**

Draw the line across the whole range — stopping short loses the mark.

**Check yourself**

calculator first · answers on page 15

1.1

Dr Eslam

Complete the table for
 $y = 2x - 1$,
 $x = -1, 0, 1, 2, 3$.

Answer:

1.2

Practice · Easy

Complete the table for
 $y = 2x + 5$,
 $x = -2, -1, 0, 1, 2$.

Answer:

1.3

Practice · Easy

$y = 8 - 2x$ for $-1 \leq x \leq 4$.
Give the end points of
the line.

Answer:



VARIANT 2

VARIANT 2

Read m and c

Make y the subject; then m is the number with x , c is the number on its own.

SEEN IN PAST PAPERS

0/7

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 8



Your notes

What is it?

In plain words

Only $y = mx + c$ shows m and c directly. $4x + 2y = 6$ must first become $y = -2x + 3$.

y-intercept: put $x = 0$. x-intercept:
put $y = 0$.

IT LOOKS LIKE

$$4x + 2y = 6$$

$$3x - 2y = 4$$

crosses the x-axis

$$4x + 2y = 6$$

$$2y = -4x + 6$$

$$y = -2x + 3$$

$$m = -2$$

$c = 3$

How do I do it?

1 Rearrange

$y = \dots$ — divide every term.

2 m

The coefficient of x (with its sign).

3 C

The number on its own: crosses at $(0, c)$.

4 x-axis

Put $y = 0$ and solve.

ON ITS OWN FIRST

Divide every term, keep the signs.

V2 Read m and c

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Find the gradient of $4x + 2y = 6$ " → $y = -2x + 3$; $m = -2$

"Where does the line cross the y -axis?" → $(0, c)$

"Show that the lines are parallel" → **Rearrange both; same m**

**WORKED EXAMPLE** DR ESLAMRead m and c · 2 marks

Find the gradient and the y -intercept of the line $4x + 2y = 6$.

STEP 1 $2y = -4x + 6$ ◀ x term across

STEP 2 $y = -2x + 3$ ◀ divide every term by 2

STEP 3 $m = -2$, $c = 3$ ◀ read

ANSWER gradient -2 , crosses at $(0, 3)$

✓ $x = 0$: $2y = 6$, $y = 3$ ✓

**WATCH OUT**

✗ $4x + 2y = 6$: $m = 4$

✓ $m = -2$

y is not on its own, so 4 is not the gradient.

**EXAM TIP**

"Parallel?" — rearrange both lines and write both gradients side by side.

**Check yourself**

calculator first · answers on page 15

2.1

Dr Eslam

Find the gradient and y -intercept of $3x - 2y = 4$.

Answer:

2.2

Dr Eslam

Where does $y = 4x - 8$ cross the x -axis?

Answer:

2.3

Practice · Medium

Are $3y = 9x - 5$ and $y = 3x + 2$ parallel?

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

●●●●●●● 3/7

PRACTICE QUESTIONS: 11

Equation from a Graph or Two Points

Read c where it crosses the y -axis; rise over run for m ; write $y = mx + c$.

Came up in: Jun 2022 P1HR Q10 • Nov 2023 P2H Q9 • May 2024 P1HR Q9

 Your notes

? What is it?

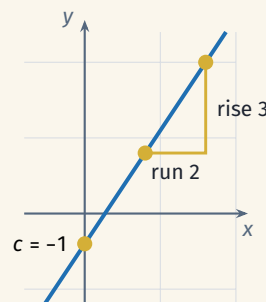
In plain words

From a graph: c is where the line crosses the y -axis. Pick two points **exactly on grid corners**; $m = \frac{\text{rise}}{\text{run}}$.

From two points: m first, then c (or $y - y_1 = m(x - x_1)$).

IT LOOKS LIKE

line on a grid (0, -3) and (2, 0)



How do I do it?

1 c Where it crosses the y -axis.**2** Two corners

Points exactly on the grid.

3 m

Rise over run; downhill is negative.

4 Write $y = mx + c$; check a third point.

c FIRST, THEN RISE OVER RUN Use the axis scales, not squares.

V3 Equation from a Graph or Two Points

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

"Line L is drawn on the grid. Find an equation for L"

→ **c from the axis; rise/run**

"passes through (0, -3) and (2, 0)"

→ $c = -3; m = \frac{3}{2}$

"in the form $y = mx + c$ "

→ **Fractions allowed:** $y = -\frac{1}{2}x + 1$ **WORKED EXAMPLE** **PAST PAPER**

June 2022 · Paper 1HR · Q10 · 3 marks

Line L is drawn on the grid. It crosses the y-axis at (0, -1) and passes through (2, 2) and (4, 5). Find an equation for L in the form $y = mx + c$.

STEP 1 $c = -1$

◀ crosses the y-axis

STEP 2 $(2, 2) \rightarrow (4, 5)$

◀ two grid corners

STEP 3 $m = \frac{5-2}{4-2} = \frac{3}{2}$

◀ rise over run

STEP 4 $y = 1.5x - 1$

◀ write it

ANSWER $y = 1.5x - 1$ ✓ (4, 5): $6 - 1 = 5$ ✓ **WATCH OUT**

× counting squares when the scale is 2

✓ use the axis numbers

The scale decides the rise and run, not the squares.

 EXAM TIP

A line going down left to right must have a negative m — check the sign.

 Check yourself

calculator first · answers on page 15

3.1

May 2024 P1HR

Line L crosses the y-axis at (0, 1) and the x-axis at (2, 0). Find $y = mx + c$.

Answer:

3.2

Nov 2023 P2H

A line passes through (0, -3) and (2, 0). Find an equation of the line.

Answer:

3.3

Dr Eslam

Find the equation of the line through (2, 5) and (6, 13).

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

●●●●●●●● **3/7**

PRACTICE QUESTIONS: 18

Parallel & Perpendicular Lines

Parallel: same m . Perpendicular: $-\frac{1}{m}$. Then use the point to find c .

Came up in: Jun 2024 P2H Q18 • Nov 2024 P2H Q18 • Jun 2025 P2HR Q10

 Your notes

? What is it?

In plain words

Get the gradient of the given line (**y alone first**). Parallel: **same**.

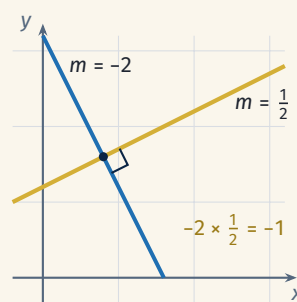
Perpendicular: **flip and change sign**.

Put the given point into $y = mx + c$ to find c .

IT LOOKS LIKE

⊥ $2x + y = 9$ through $(8, 11)$

|| $y = 2 - 5x$



How do I do it?

1 Given m

Rearrange the given line.

2 New m

Same, or $-\frac{1}{m}$.

3 Find c

Substitute the point.

4 Write

In the form asked.

FLIP AND CHANGE SIGN $-2 \rightarrow \frac{1}{2}$, $\frac{2}{5} \rightarrow -\frac{5}{2}$.

V4 Parallel & Perpendicular Lines

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“parallel to $y = 2 - 5x$... passes through $(0, 6)$ ”

→ $m = -5, c = 6$

“perpendicular to $2x + y = 9$ ”

→ $y = -2x + 9$: **new** $m = \frac{1}{2}$

“Find the gradient of a line perpendicular to P”

→ **Just** $-\frac{1}{m}$ — **no equation needed**

WORKED EXAMPLE PAST PAPER

November 2024 · Paper 2H · Q18 · 4 marks

The straight line **L** is perpendicular to $2x + y = 9$ and passes through $(8, 11)$. Find an equation for **L** in the form $y = mx + c$.

STEP 1 $y = -2x + 9 \Rightarrow m = -2$

◀ given line, y alone

STEP 2 $m_L = \frac{1}{2}$

◀ flip and change sign

STEP 3 $11 = \frac{1}{2}(8) + c \Rightarrow c = 7$

◀ use the point

STEP 4 $y = \frac{1}{2}x + 7$

◀ write

ANSWER $y = \frac{1}{2}x + 7$

✓ $-2 \times \frac{1}{2} = -1$ ✓

WATCH OUT

× $2x + y = 9$: $m = 2$

✓ $m = -2$

Rearrange first: $y = -2x + 9$.

EXAM TIP

Check perpendicular gradients multiply to -1 .

Check yourself

calculator first · answers on page 15

4.1

Jun 2024 P2H

P: $5y + 2x = 7$. Find the gradient of a line perpendicular to **P**.

Answer:

4.2

Jun 2025 P2HR

L is parallel to $y = 2 - 5x$ and passes through $(0, 6)$. Find **L**.

Answer:

4.3

Dr Eslam

Find the line perpendicular to $y = 3x - 1$ through $(6, 2)$.

Answer:

V5

VARIANT 5

Lines and Shapes

Intercepts give triangle corners; two lines meet where their equations agree.

SEEN IN PAST PAPERS

0/7

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 14



Your notes

? What is it?

In plain words

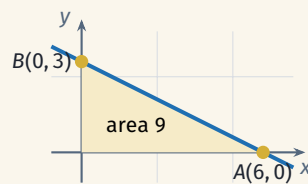
Where a line crosses the **axes**: put $x = 0$, then $y = 0$. With the origin these make a **right-angled triangle**:
area = $\frac{1}{2} \times \text{base} \times \text{height}$.

Two lines **meet** where both equations are true: set them equal.

IT LOOKS LIKE

area of AOB

$y = x + 1$ meets $y = 5 - x$



How do I do it?

1 Sketch

Draw the lines roughly; label points.

2 Intercepts

$x = 0$ and $y = 0$.

3 Meet

Set the y 's equal, solve for x , then y .

4 Measure

Area, length or midpoint as asked.

SET THE y's EQUAL

SET THE y's EQUAL Intercepts: $x = 0, y = 0$.

V5 Lines and Shapes

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

"The line crosses the axes at A and B.
Find the area of AOB"

→ **Intercepts; $\frac{1}{2} \times \text{base} \times \text{height}$**

"The lines meet at P"

→ **Set equal; solve**

"perpendicular bisector of AB"

→ **Midpoint, $-\frac{1}{m}$ (Topic 2)** **WORKED EXAMPLE** DR ESLAM

Lines and shapes · 3 marks

The line $y = -\frac{1}{2}x + 3$ crosses the x-axis at A and the y-axis at B. O is the origin. Find the area of triangle AOB.

STEP 1 $x = 0 : B(0, 3)$ ◀ y-intercept

STEP 2 $0 = -\frac{1}{2}x + 3 \Rightarrow A(6, 0)$ ◀ x-intercept

STEP 3 $\frac{1}{2} \times 6 \times 3$ ◀ right angle at O

STEP 4 $= 9$ ◀ area

ANSWER 9 square units

✓ base 6, height 3 ✓

**WATCH OUT**

× x-intercept: $x = -\frac{1}{2} \times 0 + 3$
✓ put $y = 0$

The x-axis is where $y = 0$.

**EXAM TIP**

A sketch with the intercepts marked earns method marks on its own.

**Check yourself**

calculator first · answers on page 15

5.1

Dr Eslam

The lines $y = x + 1$ and $y = 5 - x$ meet at P. Find P.

Answer:

5.2

Dr Eslam

Find the area of the triangle formed by $y = 2x$, $y = 6 - x$ and the x-axis.

Answer:

5.3

Dr Eslam

A(1, 2), B(5, 4). Find the perpendicular bisector of AB.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $-3, -1, 1, 3, 5$

1.2 $1, 3, 5, 7, 9$

1.3 $(-1, 10)$ and $(4, 0)$

2.1 $m = 1.5, (0, -2)$

2.2 $(2, 0)$

2.3 Yes: both $m = 3$

3.1 $y = -\frac{1}{2}x + 1$

3.2 $y = 1.5x - 3$

3.3 $y = 2x + 1$

4.1 $\frac{5}{2}$

4.2 $y = 6 - 5x$

4.3 $y = -\frac{1}{3}x + 4$

5.1 $P(2, 3)$

5.2 12

5.3 $y = -2x + 9$

THE WHOLE TOPIC ON ONE PAGE

Linear graphs cheat sheet



Your notes
V1 Draw 1/7

$$y = 7 - 4x$$

Table, plot, rule.

V2 Read m, c

$$4x + 2y = 6 \rightarrow y = -2x + 3$$

 y alone first.**V3 From a graph 3/7**

$$y = 1.5x - 1$$

 c first, then rise over run.**V4 Parallel, $\perp$ 3/7**

$$-2 \rightarrow \frac{1}{2}$$

Same m or $-\frac{1}{m}$; point gives c .**V5 Lines & shapes**

$$\frac{1}{2} \times 6 \times 3 = 9$$

Intercepts; set the y 's equal.**TOP 3 LOST MARKS**

1. reading m before y is alone $\times$
2. counting squares, not the scale $\times$
3. perpendicular: only the sign changed $\times$

BOTTOM LINE $y = mx + c$ · y alone first · rise over run · parallel same, perpendicular $-\frac{1}{m}$.